Who is who: injury rates between players and within each player in professional team sports

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Abstract

Objective Per-hour injury rates guide rest and match calendars. We measured how injury rates differ between players and with recent load.
Methods Observational study of public match records from nine team sports, both sexes. Injury records covered 22 573 players in six sports (748 women, in one league). In football, injuries were injury-coded substitutions. A heavy week had a full match or more within seven days; a longer-gap week followed an 8-to-13-day gap. We compared heavy weeks across players and within each player.
Results As expected, players' usual injury rates differed: between two players picked at random, the higher was typically 1.83 times the lower. As known, two or more injuries of 28 days or more in three years went with 1.3 to 1.7 times the rate in five sports. Most of the difference between players lay outside that history. As expected, within the same player, playing time rose less after a heavy week than across players, whether or not a replaced player may return. As expected, simulations with no effect of load read heavy weeks as riskier within each player's season in all six sports, because injuries lower later load.
Conclusion Per-hour rates that compare players mix recent load with who carries it. Researchers and injury surveillance programmes who compare each player's heavy weeks with the same player's longer-gap weeks should first check that comparison by simulation. Club medical staff can read lower injury rates among their most-used players as a sign of who plays.

What is already known on this topic

- Players differ in injury risk beyond their load and their previous injuries; those who play most can look safer, because players at lower risk play more.
- Per-hour injury rates assume that the hours a player records do not depend on the load under study.

What this study adds

- In every sport whose records could measure it, players' usual injury rates differed widely; because an injury lowers later load, simulations with no effect of load read heavy weeks as riskier within each player's season.

How this study might affect research, practice or policy

- Researchers and injury surveillance programmes should check by simulation any comparison of a player's heavy weeks with the same player's longer-gap weeks and, where they infer injury from team sheets, should not read a missed start in the next match as injury.
- Club medical staff can read a lower injury rate among their most-used players as a sign of who plays; they can read repeated injuries of 28 days or more as a known marker of higher risk.

Introduction

Injury rates guide which players medical teams rest and how many matches players play.[1, 2] Both uses assume that recent load, meaning last week's play, drives injury risk. Yet some players are injured more often than others. A previous injury, for example, is a strong risk factor for the next one.[3] Suppose the players who play the most are also those injured least. Then injury rates that compare players will make a heavy week, a full match or more in the previous week, look safe.

In football, injuries are counted per 1000 hours of play.[4–6] Studies of fixture congestion and recent load use such rates.[7–12] All of them assume that the hours a player records stay the same, whatever the recent load. But players who played more in the previous week play more minutes in the next match.[13, 14] We call this rise the workload gradient.

A per-hour rate divides injuries by the minutes played. The number we divide by is called the denominator. After a heavy week the minutes rise, so the denominator rises too. The rise could have two causes. A heavy week might change how long a player plays. Or the players with a heavy week might be the regular starters, those the coach picks to start every week and keeps on the field. We call this choice team selection. Which cause holds decides whether a per-hour rate compares loads or compares players.

Earlier work shows that players differ in injury risk beyond their load and their previous injuries.[12, 15, 16] In basketball, coaches rest players when their injury risk rises.[17] In medicine, comparing groups of people who were picked in some way can mislead.[18] But to our knowledge, only the first author's nine-sport preprint[19] has measured how much of the recorded playing time comes from team selection. Others have raised the question.[1, 20]

In nine team sports, we asked how much players differ in their usual injury rate. We then asked whether comparing different players makes heavy weeks look safer than comparing each player with their own weeks.

Methods

Design and data

This observational study used public match records from nine team sports: it studied matches as they were played. The records show each team's line-up and the minute of each substitution. The study follows the STROBE reporting guideline.[21] Table 1 lists the sports and names the leagues (Supplementary Section S2). The records of men's football come from Transfermarkt, which we call the provider. We asked the same questions of 37 more leagues in seven sports (Supplementary Section S9). A league sample is one league's seasons under one substitution limit, since most football leagues moved from three substitutions per match to five in 2020.[22]

Recent match load

Recent match load, or recent load for short, is the league minutes a player played in the previous seven days. It is counted in full matches of 90 minutes. A player with a full match or more was loaded: a heavy week. A player with no minutes was rested. A longer-gap week is a rested week: the club's previous league match came 8 to 13 days earlier. Then the fixture list set the rest, rather than the coach or an injury (Figure 1C).

Injury records

A substitution is injury-coded when the match report gives injury as its reason. We used only league-seasons that allowed five substitutions, and in which at least 90 per cent of substitutions have a reason. A recorded absence after an injury-coded substitution is a sign of real injury. So we also counted only the injury-coded substitutions confirmed by a recorded absence. We call this second count the narrower outcome. Football's comparisons of heavy weeks are given on both outcomes (Supplementary Section S3). In the other sports, an injury is each league's own record of an absence (Supplementary Section S12).

A statistical model gave each player their own usual injury rate. We call it the baseline rate. It takes in everything that stays the same about a player, such as position, usual load and how the club records injuries. We ran the model in each of 11 player groups (Table 1; Supplementary Section S11). In five sports, the records also give each player's injury history (Supplementary Section S12).

The injury comparison

We counted injury-coded substitutions per 1000 hours played as a starter. We call these hours starter hours. We compared heavy weeks with longer-gap weeks in two ways. Across starters, we compared different starters' weeks within each league-season and combined the league-seasons

(Mantel–Haenszel). Within each starter, we compared each starter's heavy weeks with the same starter's own longer-gap weeks in the same league-season (conditional Poisson regression). Supplementary Section S6 adds other counts and checks. We call a match completed when the starter played all 90 recorded minutes. We also made both comparisons against longer-gap weeks after a completed match.

Who plays after a heavy week

We split the workload gradient exactly into two parts: one comes from who starts, the other from starters and substitutes playing longer.[23] The within-player slope is the gradient measured within each starter. We also modelled how the chance that a starter was taken off changed with recent load (Supplementary Section S3). We ran this model in the eight European men's leagues and the other leagues that timed substitutions. It also ran in a worldwide survey of first-tier men's leagues (Table 1; Supplementary Sections S8 and S9).

Statistical analysis and pre-set criteria

The statistics follow the CHAMP statement.[24] We set two criteria before testing them (Supplementary Section S4). First, from sample to sample, the within-player slope would vary at most half as much as the rise across starters. Second, the more firmly a league keeps loaded starters on, the more of the starters' rise would come from differences between starters. Every other analysis, the injury comparison among them, is exploratory: it finds leads to check. But we chose the comparison within each starter and season after seeing the results of similar comparisons. We added the narrower outcome and the reference after a completed match later. A check had already given that reference's estimate. A simulation kept the real calendars and appearances and gave load no effect (Supplementary Sections S6 and S9). Only comparisons that it read within two per cent of one, with 95% intervals that held one at least 90% of the time, carry a sentence here. Comparisons that missed this are printed beside their simulated values. Each result is reported on its own, as if it were the only comparison. We wrote the analysis scripts with an AI assistant and checked them against hand-worked tests.

How to read the results

Most results are ratios of two injury rates. A ratio of 0.88 means a rate 12% lower than the rate it is compared with. Each ratio comes with a range, its 95% confidence interval (95% CI). The true value is likely to be in this range. When the whole range is above one, or below one, the difference is clear. When the range includes one, there is no clear difference: the two rates are similar. For heavy weeks, we call a clear rise riskier and a clear fall safer.

Equity, diversity, and inclusion statement

The league samples include women's football, Australian rules, basketball and cricket. The worldwide survey covers men's leagues only. Women's injury records come from one league, the WNBA: 748 of the 22 573 players with injury records. The two authors work at one European university, in statistics and medicine.

Patient and public involvement

Patients or the public were not involved in the design, conduct, reporting or dissemination plans of this research.

Table 1: Which data answer which question. The nine sports are football, rugby union, Australian rules, basketball, ice hockey, American football, rugby league, baseball and cricket. The eight European men's leagues are those of England, France, Germany, Italy, the Netherlands, Portugal, Spain and Turkey. League-seasons, players and sizes are totals over the samples. In the first rows, the league samples are the 11 player groups. A player in two samples of one question counts twice, unless the source gives each player one identifier: the NBA with the G League, and MLB's batters with its pitchers. The worldwide survey, the injury comparison and the next start count each player once. The survey's pooled results combine the 140 samples played with three or five substitutions allowed (Supplementary Table S33). One start in 31 ended in an injury-coded substitution (Supplementary Section S3). The next start asks how often a starter taken off early started the club's next match (Supplementary Section S3). Outside football, an injury is each league's own record, counted once per absence (Supplementary Section S12). Median age is the range of the samples' median ages at appearance; a dash marks rows without one (Supplementary Table S2). The injury comparison's row includes goalkeepers, whom Figure 1C and Figure 2 leave out (19 231 starts).

Question	Sport and sex	League samples	League-seasons	Players	Of them, women	Median age, years	Size	Unit of size
Players against recent load (Figure 1A) and injury history (Figure 1B)	Football, men: England, France, Germany, Italy and Spain, as one group	1	27	4591	0	–	6710	injury-coded substitutions, outfield starters
	Australian rules, men	1	9	1253	0	–	1824	absences through injury
	Basketball, men and women	4	62	4252	748	–	3180	absences through injury; injured-list placements in Japan
	Ice hockey, men	2	13	1932	0	–	1946	absences through injury
	Baseball, men, batters and pitchers	2	32	3965	0	–	4743	injured-list placements
	American football, men	1	13	6580	0	–	4613	absences through injury
Who starts (Figure S3A)	Football, 10 men's and 7 women's; rugby union, men	18	74	18 419	2104	24.5–28.7	685 148	appearances
Within the same player (Figure S3B)	Football, men and women; rugby union, men; Australian rules, women	25	57	18 673	3265	24.4–28.5	14 963	players with two or more starts
Kept on, worldwide survey	Football, men's first tiers in 96 of 99 competitions, the eight European men's leagues fitted apart	140 of 162	538	72 208	0	–	2 089 384	starter appearances
The injury comparison (Figures 1C and 2) and one start in 31	Football, men, the same five leagues	5	27	5942	0	25.4–27.4	296 029	appearances, goalkeepers included, with 7158 injury-coded substitutions
The next start	Football, men, 14 leagues, five substitutions	14	82	13 202	0	24.8–28.7	490 125	starts followed by another club match

Results

Players differed widely in their usual injury rate

As expected, players differed in their usual injury rate. Between two players picked at random from one season, the higher baseline rate was typically 1.83 times the lower, over all 11 player groups (Figure 1A). Allowing for each team changed it little. In football it was larger on injuries confirmed by a recorded absence (1.94).

As expected, injury history also marked higher risk. Players with two or more injuries of 28 days or more in the previous three years had 1.3 to 1.7 times the injury rate of the others. This held in every sport with injury histories (Figure 1B; Supplementary Section S12). Yet, as at one Australian football club,[15] most of the difference between two players lay outside that history. As expected for a marker this modest or rare, taking it into account made that difference at most 3.1% smaller (Supplementary Table S45). Even the fullest injury record of each player made that difference at most 10.4% smaller (Supplementary Section S12).

Compared like with like, heavy weeks had similar injury rates

In European men's football, the standard comparison made heavy weeks look safe. Across starters, heavy weeks looked 12% safer than longer-gap weeks (0.88, 95% CI 0.82 to 0.94; Figure 2). In a simulation with no effect of load, this comparison read 0.97 to 0.99. It read 0.92 when players at lower risk carried more load (Supplementary Section S6). Starters with heavy weeks in more of their earlier starts had lower rates, from 30 to 14 injury-coded substitutions per 1000 starter hours.

Of heavy weeks, 95% followed a match the starter completed, against 48% of longer-gap weeks. Longer-gap weeks after a match the starter did not complete had more injury-coded substitutions (1.33, 95% CI 1.20 to 1.47). With no effect of load, a simulation gave 1.04 to 1.18. When both kinds of week followed a completed match, their injury rates were similar. Across starters they read 1.02 (95% CI 0.93 to 1.11). Within each starter and season they read 0.99 (0.91 to 1.09). With a recorded absence, they read 0.96 and 0.95. In the simulation, only this heavy-week comparison within each starter read about one (Supplementary Section S6).

Every other comparison of heavy weeks within each player and season read them as riskier in the simulation. They read 1.06 to 1.17 times in football, and 1.05 to 8.20 times in the other sports with injury records (Supplementary Sections S6 and S9). After an injury, a player misses games and comes back rested. So within a season, the weeks after an injury are rarely heavy.

Heavy weeks fell to regular starters

As the first author's nine-sport preprint reported,[19] the extra minutes after a heavy week came mostly from who starts. By an exact split, who starts accounted for 73% to 92% of the workload gradient in all 18 samples of football and rugby union (Supplementary Figure S3A). Within each starter, recorded minutes changed little. In all 54 groups of the 37 other leagues, playing time rose less within the same player than across players. As expected, starters with a full match in the previous week stayed on the field longer. This mostly reflects their roles. The pattern held in football's first tiers, in all 15 other leagues that timed substitutions, and within the same starter (Supplementary Sections S8 and S9; Section S6). Where the fixture list set a short gap, the same starter came off slightly sooner (1.06, 95% CI 1.01 to 1.12).

A summary of our key findings is presented in Figure 2.

Discussion

In football, comparing different starters made heavy weeks look safer. In every sport with injury records, simulations where load did nothing read heavy weeks as riskier within each player's season. Compared like with like, football's heavy weeks carried similar rates.

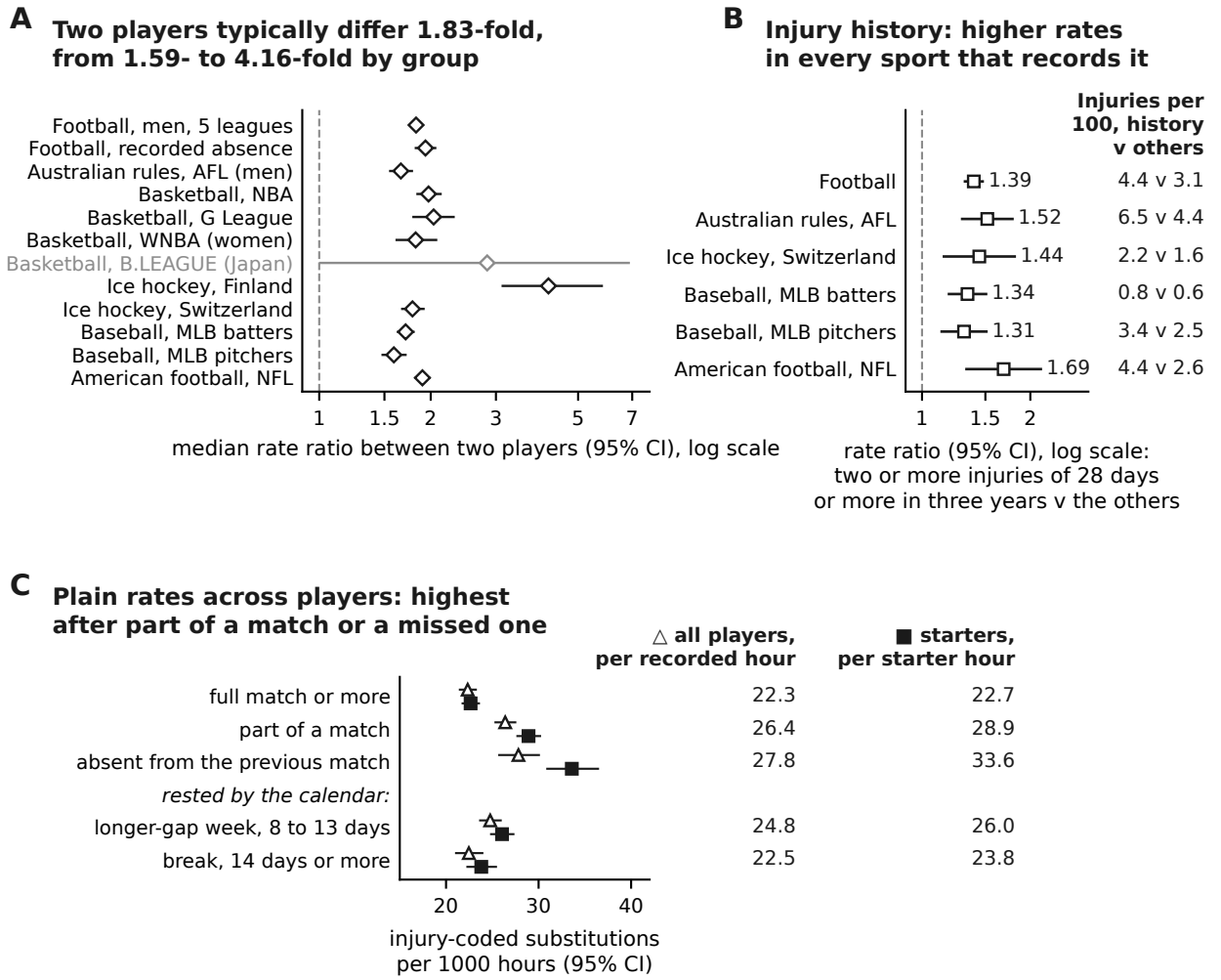


Figure 1: Players differed widely in their usual injury rate. (A) For each player group, the diamond shows how far apart two players picked at random typically are: the median rate ratio, the higher rate over the lower. Lines show 95% CIs (Supplementary Table S42). On the log scale, doubling and halving sit the same distance from one. Football counts outfield starts only, once for injury-coded substitutions and once for those a recorded absence followed. The G League leaves out absences that were likely moves to the NBA. Japan's B.LEAGUE, drawn lighter, had only 27 injuries. Finland's ice hockey league, the Liiga, has the highest ratio between players, because its teams record injuries in different ways (Supplementary Section S12). Outside football, some injury lists record rest, or a move to another team, as an injury, which would move its diamond to the right. (B) Players with two or more injuries of 28 days or more in the previous three years, against the others. The panel gives the rate ratio, and beside it each group's injuries per 100 starts or appearances (Supplementary Table S44). (C) Plain injury rates, before any comparison within players: injury-coded substitutions per 1000 hours among outfield players in the 27 league-seasons of the injury comparison, with 95% CIs. Triangles count all players per hour played, and squares count starters per starter hour. Part of a match is some minutes, but less than a full match. A player was rested by the calendar when the club had no league match that week, and absent when they missed the club's previous match.

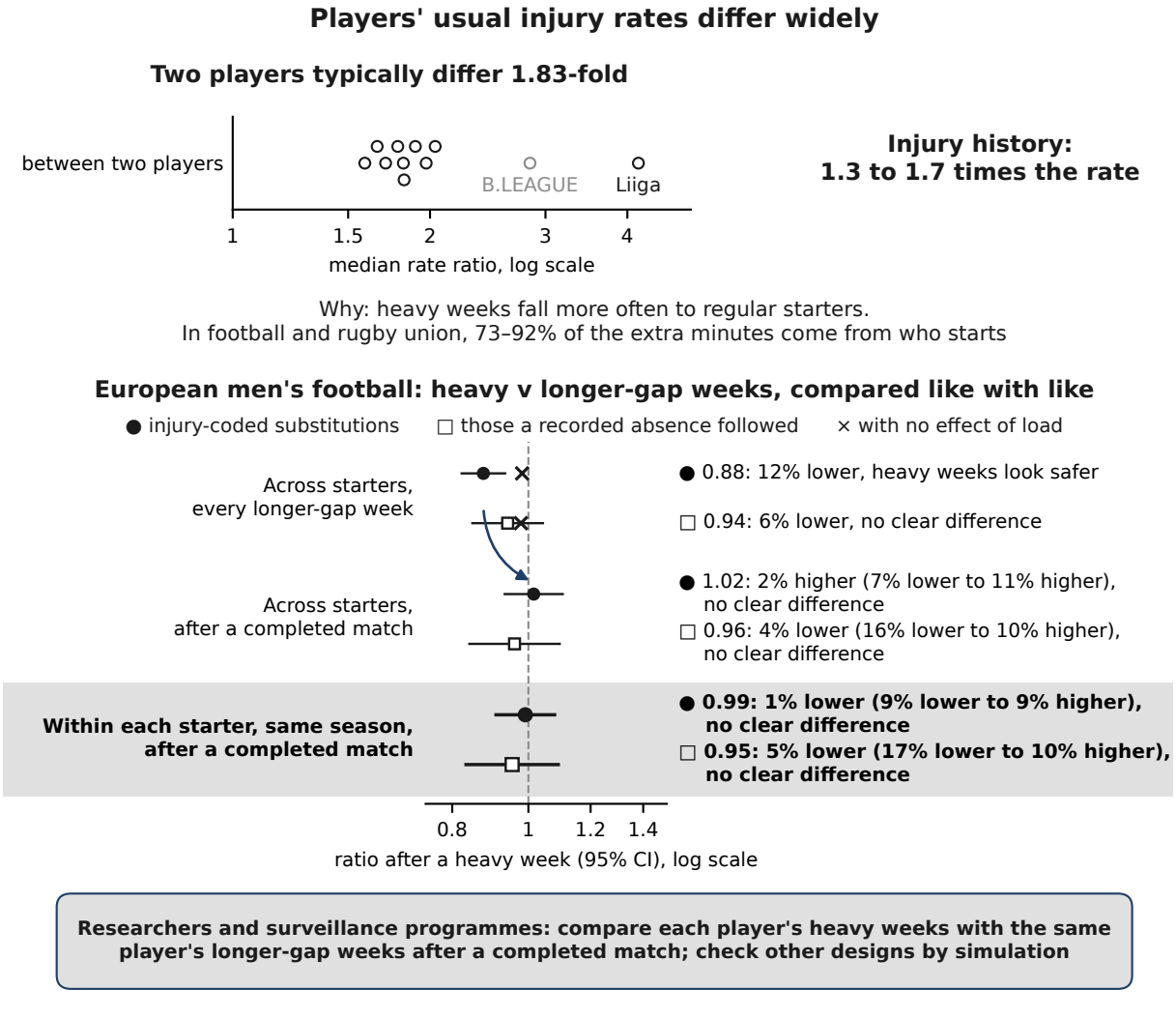


Figure 2: Central Figure: players differ widely in their usual injury rate, and how to compare like with like. In the top row, each dot is one of the 11 player groups and gives the median rate ratio between two players picked at random (Supplementary Table S42). The dot at the far right is Finland's Liiga. Its teams record injuries in different ways. Japan's B.LEAGUE, drawn lighter, had few injuries. The injury history's ratios span the six groups that record it. In the second row the symbols change meaning. A filled mark shows outfield starters' injury-coded substitutions per starter hour. An open mark shows those a recorded absence followed. The first pair is the standard comparison, across starters against every longer-gap week. A cross marks its value in a simulation with no effect of load. The other pairs use only longer-gap weeks after a completed match: across starters, and within each starter (shaded) in the same league-season. The words beside each interval give its reading: safer or riskier when it lies wholly below or above one, no clear difference when it holds one. The arrow runs from the standard comparison to the comparisons after a completed match. A heavy week had a full league match or more in the previous seven days. A longer-gap week followed a gap of 8 to 13 days. The box gives this study's own advice.

Per-hour rates divide by who starts

A per-hour rate compares different players, as well as different loads.[25] For a regular starter, a heavy week is the usual week. So most heavy weeks belong to regular starters. Workplace health research calls this the healthy worker effect: the workers who do most of a job are those fit enough to stay in it.[26–28] A basketball preprint reported the same effect for recent minutes.[29]

Other counts have been proposed: only the players fit to play,[30] or appearances rather than hours.[31–35] Counting appearances does not remove the selection among starters (Supplementary Section S6). Our earlier preprint measured how much minutes rise with load.[13] It did not compare each player with themselves.

How comparing players misleads

- **Heavy weeks look safe.** Comparing starters with each other gives the heavy week the credit for the regular starters' lower risk. This matches a healthy worker effect. Players picked for a World Cup showed the same pattern.[36] Regular match play could also protect over months.
- **As known, the denominator can flip the answer.** Loaded players had 26% more injury-coded substitutions per appearance than rested players, but 9% fewer per hour played (Supplementary Section S6). With no effect of load, a simulation read these comparisons as 1.34 and 0.97. Youth American football showed the same flip.[37]
- **Comparing players by their load misses those at higher risk,** such as players with repeated injuries of 28 days or more.

Other factors in resting a player

Besides recent load, a decision to rest a player can weigh:

- previous injury, a strong risk factor;[3]
- recovery time: among rested starters, rates were lowest after breaks of 14 days or more;
- every match played, and the travel between them;[10]
- training load, which match records never show;[11]
- squad depth and rotation.[8, 38]

Implications for research

First, compare each player with themselves only in a design that a simulation with no effect of load reads correctly. Self-controlled designs assume that an event does not change later exposure.[39] An injury does change a player's next weeks of load. In football, setting each starter's heavy weeks against the same starter's longer-gap weeks after a completed match, in the same season, passed this check. Outside football, no reference weeks that we tried passed (Supplementary Section S9). Such designs have been used on public football records before.[40] They suit surveillance programmes and research that pool clubs. Report them beside the standard comparison across players.

Second, count injuries confirmed by a recorded injury absence. Also count load over every match played (Supplementary Section S6).

Third, researchers who infer injury from team sheets should not read a missed start as injury (Supplementary Section S3).[41]

Clinical implications

First, heavy weeks looked safer when players were compared with each other. Starters with the most heavy weeks had the lowest rates. So lower per-hour injury rates among a club's most-used players can show who plays, not that their load is safe.

Second, an injury lowers a player's load in the weeks that follow. So a player's own heavy weeks can look riskier when load does nothing. In football, compared like with like, heavy and longer-gap weeks carried similar rates (0.99, 95% CI 0.91 to 1.09). The interval allows a rise of up to 9%.

Third, most of the difference between players lay outside their injury history. Two or more injuries of 28 days or more in three years mark a group at higher risk, not the players who will be injured.[42] Clubs already rank previous injury among their main risk factors.[43] The history confirms a known risk factor and should not by itself change selection (Supplementary Section S12).

Limitations

The most important limitation is how recent match load was measured.[44] The main analysis counts league matches only. Matches in the international breaks are never counted. So an international player's load is understated in exactly the weeks that congestion research studies (Supplementary Table S12).

Second, the records are public match data, not surveillance data kept by club doctors.[45, 46] The injury reason is a label from the match report, not a diagnosis. It also covers players taken off as a precaution. A recorded absence followed 41% of heavy weeks' injury-coded substitutions, against 38% after longer gaps (Supplementary Section S6). No video audit of the study seasons was feasible. Recorded minutes leave out added time. Giving it back changed the per-hour ratios little (Supplementary Sections S3 and S6).

Third, the design is observational: it records what happened and changes nothing.[47] Selection from week to week can still bias the result. For example, a player who came off early with a knock may carry more risk into the next start. An injury's effect on later load also biases comparisons within players. Leaving out the weeks after each injury did not remove this bias (Supplementary Section S6). So the only heavy-week comparison of injury rates within players that we read is the one after a completed match. In football and rugby union, recorded minutes stop at the full match. This sets the smallest within-player slope the design could detect (Supplementary Section S6).

Fourth, we saved the two pre-set criteria, with a date, to the project's own files, not to a public protocol. A public protocol is a plan published in advance. We saved both after our earlier preprint had used the same records. The first passed narrowly and the second failed. On the first, the within-player slope varied 0.48 times as much as the rise across starters, against a limit of one half. The second failed because its link was too uncertain to tell either way (Supplementary Section S4). We also registered a confirmation on the next complete seasons, but the study ends before those seasons do.

Finally, the samples differ in sport, sex, era and size. The only women's injury record is one basketball league. The injury comparison comes from European men's football (Supplementary Section S9). Outside football, an injury is dated to a game the player played (Supplementary Sections S9 and S11).

Conclusion

Players' usual injury rates differ widely. Heavy weeks fall more often to regular starters. So per-hour rates that compare players mix recent load with who carries it. An injury also lowers a player's later load. This biases comparisons within players. In European men's football, heavy and longer-gap weeks carried similar rates when both followed a completed match. Researchers and injury surveillance programmes should check comparisons within players by simulation. In football, they should compare each player's heavy weeks with the same player's longer-gap weeks after a completed match.

Declarations

Contributors GPR designed the study, collected and analysed the data and wrote the manuscript. NM shaped the methods and the clinical framing of the earlier preprint that this manuscript extends. Both authors approved the final version. GPR is the guarantor. Anthropic's Claude, a generative AI assistant, was used under GPR's direction to write and check analysis code, search the literature and edit the text. It was used because the first author could not otherwise build and check analyses of this size. The study design, the claims and their interpretation are the authors'. Every reported quantity is computed from the committed data by the archived code and was verified by GPR. Both authors take full responsibility for the content.

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Competing interests None declared.

Prior dissemination This manuscript is posted on arXiv as the next version of our earlier preprint.[13] That preprint raised the question this manuscript answers. Some of this manuscript's results were first posted in the first author's nine-sport preprint.[19]

Ethics approval The study reused only public, pre-existing match records. There was no recruitment, no contact with any individual, no intervention and no access to medical records or clinical files, so informed consent did not apply. The records name professional athletes, and some reasons for substitution are injuries. They were therefore treated as special-category personal data despite being public. Processing was for scientific research under the research provisions of the General Data Protection Regulation. Everything published, every table, figure and deposited file, is aggregated to whole league samples, so none of it describes a single player. Ethical approval was not sought. These published records were treated as archival data, which Trinity College Dublin's research ethics guidance excepts from approval.[48] The same guidance grants no approval retrospectively.

Data availability statement Data are available in a public, open access repository. The aggregated results behind every table and figure, and the analysis scripts, are in the archived record at DOI 10.5281/zenodo.24194740. No player-level record is shared. The match records the scripts read identify players, so they are not redistributed. Their reconstruction from the public providers, each source's terms of use as read and any permission held are documented in the archive. Statistics from NBA.com are credited to NBA.com, as its terms ask.

References

[1] FIFPRO. Excessive workload demands: player performance, recovery and health. FIFPRO Player Workload Monitoring men's football report, 2023/24 season; 2024. With Football Benchmark; the core group of players, page 15. Accessed 28 September 2026. Available from: `https://media.fifpro.org/media/1oxkdnt2/en_pwm_men-s-annual-report-2024.pdf`.

[2] den Hollander S, Burgess D, Kerkhoffs G, et al. The International Match Calendar in men's professional football: an expert position statement. Scand J Med Sci Sports 2025;35:e70163. doi:10.1111/sms.70163.

[3] Hägglund M, Waldén M, Ekstrand J. Previous injury as a risk factor for injury in elite football: a prospective study over two consecutive seasons. Br J Sports Med 2006;40:767–72. doi:10.1136/bjsm.2006.026609.

[4] Bahr R, Clarsen B, Derman W, et al. International Olympic Committee consensus statement: methods for recording and reporting of epidemiological data on injury and illness in sport 2020 (including STROBE Extension for Sport Injury and Illness Surveillance (STROBE-SIIS)). Br J Sports Med 2020;54:372–89. doi:10.1136/bjsports-2019-101969.

[5] Waldén M, Mountjoy M, McCall A, et al. Football-specific extension of the IOC consensus statement: methods for recording and reporting of epidemiological data on injury and illness in sport 2020. Br J Sports Med 2023;57:1341–50. doi:10.1136/bjsports-2022-106405.

[6] Ekstrand J, Spreco A, Bengtsson H, et al. Injury rates decreased in men's professional football: an 18-year prospective cohort study of almost 12 000 injuries sustained during 1.8 million hours of play. Br J Sports Med 2021;55:1084–91. doi:10.1136/bjsports-2020-103159.

[7] Bengtsson H, Ekstrand J, Hägglund M. Muscle injury rates in professional football increase with fixture congestion: an 11-year follow-up of the UEFA Champions League injury study. Br J Sports Med 2013;47:743–7. doi:10.1136/bjsports-2013-092383.

[8] Carling C, McCall A, Le Gall F, et al. The impact of short periods of match congestion on injury risk and patterns in an elite football club. Br J Sports Med 2016;50:764–8. doi:10.1136/bjsports-2015-095501.

[9] Page RM, Field A, Langley B, et al. The effects of fixture congestion on injury in professional male soccer: a systematic review. Sports Med 2023;53:667–85. doi:10.1007/s40279-022-01799-5.

[10] Allen T, Taberner M, Zhilkin M, et al. Evolving running load demands and fixture congestion in the English Premier League: a decade of insights from 2015/2016 to 2024/2025. Br J Sports Med 2025;59:1643–54. doi:10.1136/bjsports-2025-110430.

[11] Impellizzeri FM, Shrier I, McLaren SJ, et al. Understanding training load as exposure and dose. Sports Med 2023;53:1667–79. doi:10.1007/s40279-023-01833-0.

[12] Hudson SJ, Williams S, Starling L, et al. Training-related concussions in men's English Premiership Rugby Union: an eight-season prospective cohort study of 153 243 player-weeks. Br J Sports Med 2026;60:1394–400. doi:10.1136/bjsports-2026-111653.

[13] Ricou GP, Mahony N. Playing time tracks recent exposure in fifteen men's and women's football leagues: the denominator gradient and what it does to per-minute injury rates; 2026. arXiv:2608.11228v3 [stat.AP]. Not peer reviewed.

[14] Mandorino M, Kavanagh R, Tessitore A, et al. "How many minutes does the player have in his legs?" Answering one of football's oldest coaching questions through a mathematical model. Appl Sci 2026;16:2139. doi:10.3390/app16042139.

[15] Esmaeili A, Hopkins WG, Stewart AM, et al. The individual and combined effects of multiple factors on the risk of soft tissue non-contact injuries in elite team sport athletes. Front Physiol 2018;9:1280. doi:10.3389/fphys.2018.01280.

[16] Lewis M. It's a hard-knock life: game load, fatigue, and injury risk in the National Basketball Association. J Athl Train 2018;53:503–9. doi:10.4085/1062-6050-243-17.

[17] Krishna K, Lychagin S, McLean LS. Injury, workload, and the strategic response of coaches in the NBA. Cambridge, MA: National Bureau of Economic Research; 2026. 35803. doi:10.3386/w35803.

[18] Griffith GJ, Morris TT, Tudball MJ, et al. Collider bias undermines our understanding of COVID-19 disease risk and severity. Nat Commun 2020;11:5749. doi:10.1038/s41467-020-19478-2.

[19] Ricou GP. One equation allocates playing time in competitive team sports: the rulebook sets the arithmetic and the league sets the habit; 2026. arXiv:2609.05584v1 [physics.soc-ph]. Not peer reviewed.

[20] McKay C, Cross M, Kemp S, et al. Strategies used by professional rugby union clubs to manage players for artificial turf exposure. S Afr J Sports Med 2020;32:1–7. doi:10.17159/2078-516x/2020/v32i1a8276.

[21] von Elm E, Altman DG, Egger M, et al. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. Lancet 2007;370:1453–7. doi:10.1016/S0140-6736(07)61602-X.

[22] The IFAB. Five substitutes option temporarily allowed for competition organisers (8 May 2020); and The IFAB permanently approves five-substitute option in top-level competitions (13 June 2022); 2022. The 2020 announcement: `https://www.theifab.com/news/five-substitutes-option-temporarily-allowed-for-competition-organisers/`. Both accessed 27 September 2026. Available from: `https://www.theifab.com/news/the-ifab-permanently-approves-five-substitute-option-in-top-level-competitions/`.

[23] Kitagawa EM. Components of a difference between two rates. J Am Stat Assoc 1955;50:1168–94. doi:10.2307/2281213.

[24] Mansournia MA, Collins GS, Nielsen RO, et al. A CHecklist for statistical Assessment of Medical Papers (the CHAMP statement): explanation and elaboration. Br J Sports Med 2021;55:1009–17. doi:10.1136/bjsports-2020-103652.

[25] Moreno-Pérez V, Carles Bové B, Courel-Ibáñez J, et al. Non-contact injury incidence in professional women's football depends on the starting status of the player. Biol Sport 2025;42:99–107. doi:10.5114/biolsport.2025.145916.

[26] Li CY, Sung FC. A review of the healthy worker effect in occupational epidemiology. Occup Med (Lond) 1999;49:225–9. doi:10.1093/occmed/49.4.225.

[27] Steenland K, Deddens J, Salvan A, et al. Negative bias in exposure-response trends in occupational studies: modeling the healthy worker survivor effect. Am J Epidemiol 1996;143:202–10. doi:10.1093/oxfordjournals.aje.a008730.

[28] Warren JR, Rumore G. The association between playing professional American football and longevity. Proc Natl Acad Sci U S A 2023;120:e2308867120. doi:10.1073/pnas.2308867120.

[29] Yu Y, Hu G. The load management paradox: correcting the healthy-worker survivor effect in NBA injury modeling; 2026. arXiv:2603.26935 [stat.AP]. Not peer reviewed. doi:10.48550/arXiv.2603.26935.

[30] El-Khoury J, Stovitz SD, Shrier I. Estimating unbiased sports injury rates: a compendium of injury rates calculated by athlete exposure and athlete at risk methods. Br J Sports Med 2021;55:354–5. doi:10.1136/bjsports-2020-102456.

[31] Knowles SB, Marshall SW, Guskiewicz KM. Issues in estimating risks and rates in sports injury research. J Athl Train 2006;41:207–15. PMID: 16791309.

[32] Stovitz SD, Shrier I. Injury rates in team sport events: tackling challenges in assessing exposure time. Br J Sports Med 2012;46:960–3. doi:10.1136/bjsports-2011-090693.

[33] Hecksteden A, Berger J, Mazzolari R, et al. How to mitigate injury risk after return-to-play in elite football? Expert judgement meets data analytics. Sci Med Footb 2026:1–16. Published online ahead of print. doi:10.1080/24733938.2026.2642654.

[34] Stovitz SD, Shrier I. Determination of athlete-exposures in team sports: it depends on the research question. Br J Sports Med 2025;59:1107–9. doi:10.1136/bjsports-2025-109822.

[35] Danielsen AC, Gompers A, Bekker S, et al. Limitations of athlete-exposures as a construct for comparisons of injury rates by gender/sex: a narrative review. Br J Sports Med 2025;59:177–84. doi:10.1136/bjsports-2024-108812.

[36] Bengtsson H, Hägglund M, Ekstrand J, et al. No major changes in injury incidence in European club football during the 2022/23 FIFA World Cup season: a subanalysis of the UEFA Elite Club Injury Study. BMJ Open Sport Exerc Med 2025;11:e002772. doi:10.1136/bmjsem-2025-002772.

[37] Kerr ZY, Yeargin SW, Djoko A, et al. Examining play counts and measurements of injury incidence in youth football. J Athl Train 2017;52:955–65. doi:10.4085/1062-6050-52.7.06.

[38] Carling C, McCall A, Le Gall F, et al. What is the extent of exposure to periods of match congestion in professional soccer players? J Sports Sci 2015;33:2116–24. doi:10.1080/02640414.2015.1091492.

[39] Petersen I, Douglas I, Whitaker H. Self controlled case series methods: an alternative to standard epidemiological study designs. BMJ 2016;354:i4515. doi:10.1136/bmj.i4515.

[40] Hampstead W, Pappas E, Whalan M, et al. Match exposure and estimated travel before ACL injury in professional women's football: an exploratory within-player case-crossover study. Sports Med Open 2026;12:170. doi:10.1186/s40798-026-01144-4.

[41] Genovesi F, Clarsen B, Bongiovanni T. Availability in elite football: a practical framework for recording and communicating player health status. Br J Sports Med 2026;60:1374–7. doi:10.1136/bjsports-2026-112239.

[42] Bahr R. Why screening tests to predict injury do not work—and probably never will...: a critical review. Br J Sports Med 2016;50:776–80. doi:10.1136/bjsports-2016-096256.

[43] McCall A, Carling C, Davison M, et al. Injury risk factors, screening tests and preventative strategies: a systematic review of the evidence that underpins the perceptions and practices of 44 football (soccer) teams from various premier leagues. Br J Sports Med 2015;49:583–9. doi:10.1136/bjsports-2014-094104.

[44] Wang C, Kaufman JS, Shrier I. Exposure misclassification: an "immortal time" bias in observational studies of training load and injury. J Clin Epidemiol 2024;176:111581. doi:10.1016/j.jclinepi.2024.111581.

[45] Hoenig T, Edouard P, Krause M, et al. Analysis of more than 20,000 injuries in European professional football by using a citizen science-based approach: an opportunity for epidemiological research? J Sci Med Sport 2022;25:300–5. doi:10.1016/j.jsams.2021.11.038.

[46] Krutsch V, Grechenig S, Loose O, et al. Injury analysis in professional soccer by means of media reports: only severe injury types show high validity. Open Access J Sports Med 2020;11:123–31. doi:10.2147/OAJSM.S251081.

[47] Shrier I, Wang C, Stokes T, et al. Causal effects, workload and injury risk: the importance of specifying the research question. J Sci Med Sport 2022;25:574–8. doi:10.1016/j.jsams.2022.03.018.

[48] Trinity College Dublin. Guidance for using the Research Ethics Application Management System (REAMS): making an application; 2022. Institutional guidance, pages 5 and 27–28. Accessed 25 September 2026. Available from: `https://www.tcd.ie/research/support/assets/pdf/Top%20Guidance_Simplified-31st%20Oct.pdf`.